Module registry

Describe what is configured. Do not turn a claim into authority.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A catalogue of equipment labels: it says which instrument is installed and what it advertises, not whether you may use it or whether every claim is proven.

Why it matters: A consumer must select semantics rather than a provider name. Explicit descriptors expose compatibility, placement and evidence gaps before effects occur.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

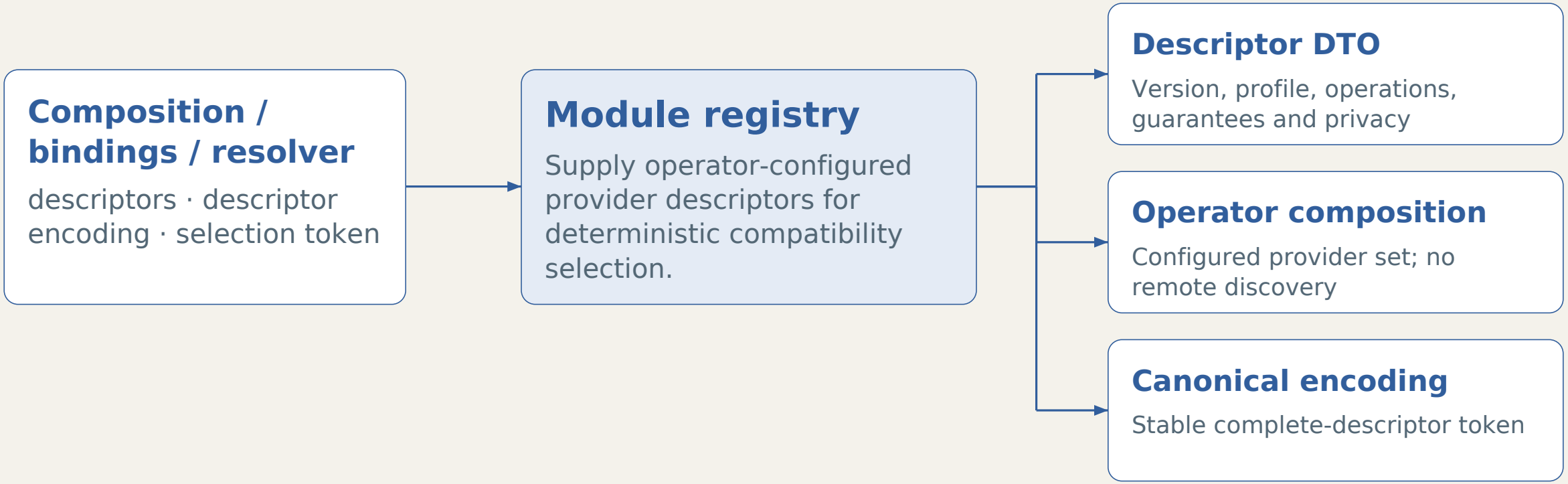
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	**Registry**
		Attention · C3	Resolver
		Inference · C4	Authorization

Supply operator-configured provider descriptors for deterministic compatibility selection.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Configured descriptor list

Provider identity, implementation version and advertised contract metadata

No runtime domain journal

No job, belief, balance, registration transaction or mutable remote catalogue

OUTSIDE THIS RESPONSIBILITY

Permission and proof

A descriptor grants no capability and proves no advertised guarantee

Dynamic discovery

No gossip, chain registration, health scheduler or public mutation API

Static catalogue responsibility · static/1; descriptor and implementation versions are separate

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
descriptors	Return configured M1 provider descriptors	Independent of transport; selected AtomStorage provider explicit
descriptor_wire / descriptor_read	Encode/decode typed metadata	Preserve supported fields and sets
selection_token	Bind the complete canonical descriptor	Configuration consistency, not remote authentication
Composition catalogues	Supply private cognition/lookup selections	Explicit profile requirements, not automatic inference

Trusted operator configuration; no independent verification of provider claims.

A worked example through the boundary

LOCAL IMPLEMENTATION · configuration and compatibility fixtures; no registry network

01 · STEP

Configure

Operator installs DeTTa-backed AtomStorage.

02 · STEP

Describe

Publish version, operations, profile and local durability.

03 · STEP

Select

Resolver matches a fixed consumer requirement.

04 · STEP

Bind

Invocation checks the complete selected token again.

What the example establishes

A useful descriptor makes both capabilities and uncertainty visible.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Description ≠ permission

Authorization remains a separate current decision.

I2

Description ≠ evidence

Advertised guarantees or measured_slo fields are not attestations.

I3

Complete identity

Changing any selected descriptor field changes the binding token.

I4

Transport neutrality

Selecting stdio vs in-process does not redefine semantics.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Operator changes configured descriptor	An existing binding fails closed on token mismatch.	No silent provider switch
Descriptor advertises unsupported feature	Resolver refuses the incompatible requirement.	Explicit compatibility error
Process restart	Re-read operator configuration; old caller intent still pins selection.	No registry state recovery transaction

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

3

M1 provider descriptors

AtomStorage, Messaging and Execution catalogue

2 × 2

Storage provider/binding matrix

Same consumer selection semantics exercised

0

Proof verifiers in registry

Required verification is not supplied by advertising

No catalogue lookup latency, update propagation or large-provider-count benchmark exists. Descriptor counts are implementation scope; conformance tests check consistency and refusal.

PROPOSED BENCHMARK · NOT RUN

Proposed: fixed valid/invalid descriptor corpora at increasing catalogue size; measure encode/decode/token costs and memory, separately from resolver selection.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Executed local descriptor construction/encoding and composition code.
FIXTURE / LOCAL TEST POLICY	Operator configuration, advertised metadata and synthetic compatibility descriptors.
BLOCKED / UNSUPPORTED	Verified remote guarantees cannot be satisfied: no proof verifier is implemented.
UNRUN	Remote discovery, dynamic registration, health monitoring, signed attestations and registry scaling.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Make metadata evidence-backed	Verifier identity and evidence expiry contract	Rejected stale or unverified claims stay visible
Improve operator inspection	Read-only catalogue and selection explanation	Show why a provider is installed versus selected
Explore dynamic catalogues	Ownership, consistency and revocation semantics first	No name-only or stale-token fallback

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

A useful descriptor makes both capabilities and uncertainty visible.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m11.py --fixture-only
```

Fixture-only validates registry/resolver/binding behavior without claiming real DA, Hive or chain. Private compositions have additional explicit descriptor checks in their own suites.

`fusion/registry.py`
`fusion/bindings.py`
`fusion/cognition.py`
`fusion/lookup_composition.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.